

Employment

Software Engineer

RelationalAI, Logic Engine team

May 2025 - Present

Remote

- Architected and implemented the imperative extension to RelationalAI's query language
 - Delivered 10x+ speedups to enterprise queries over terabytes of data
 - Collaborated across teams to engineer new implementations of graph algorithms
- Maintained the Protocol Buffer-based entry point into logic engine (~ 100k queries/day)
- Implemented user and internal observability tools to diagnose slow queries

Graduate Researcher

Northeastern University, Programming Research Lab

Sep 2022 - Apr 2025

Boston, USA

- Designed and published a programming language for optimization, beating SotA [1]
- Co-conducted and published an analysis of proof assistant community ecosystems [2]
- Assisted professors in teaching introductory classes in programming languages

R&D Intern

RelationalAI, Logic Engine team

May 2024 - Dec 2024

Remote

- Developed a reverse compiler for Datalog to increase observability into query optimizations
- Integrated visualizations of rule optimization into the profiler for our query language

REU Researcher

University of Tennessee

Jun 2021 - Jul 2021

Chattanooga, USA

- Researched and published in the theory of matrices in function spaces [3]
- Automated routine computation using Mathematica, saving weeks of in-person work

Education

Master of Science, Computer Science

Northeastern University

Sep 2022 - May 2025

Boston, USA

Bachelor of Science, Mathematics and Philosophy

Carnegie Mellon University

Aug 2018 - May 2022

Pittsburgh, USA

Publications

- [1] Minsung Cho, John Gouwar, and Steven Holtzen. "Scaling Optimization over Uncertainty via Compilation". In: *Proc. ACM Program. Lang.* 9.OOPSLA1 (2025). DOI: 10.1145/3720500.
- [2] Gwenyth Lincroft et al. "Thirty-Three Years of Mathematicians and Software Engineers: A Case Study of Domain Expertise and Participation in Proof Assistant Ecosystems". In: *21st IEEE/ACM International Conference on Mining Software Repositories, MSR 2024*. ACM, 2024, pp. 1–13. DOI: 10.1145/3643991.3644908.
- [3] Minsung Cho et al. "The Krein-von Neumann extension of a regular even order quasi-differential operator". In: *Opuscula Mathematica* 41.6 (2021), pp. 805–841. DOI: 10.7494/OpMath.2021.41.6.805.

Awards and Recognition

1st Place, Student Research Competition, Programming Language Design & Implementation (PLDI)

Scholarship Recipient, Programming Languages Mentoring Workshop at PLDI (PLMW)

University and College Honors, Carnegie Mellon University

Skills

Programming languages: Julia, Python, Datalog, OCaml, Rust, Go, C, Haskell, Standard ML

Tools: Unix, Git, Jira, Docker, Datadog, Observe, Snowflake